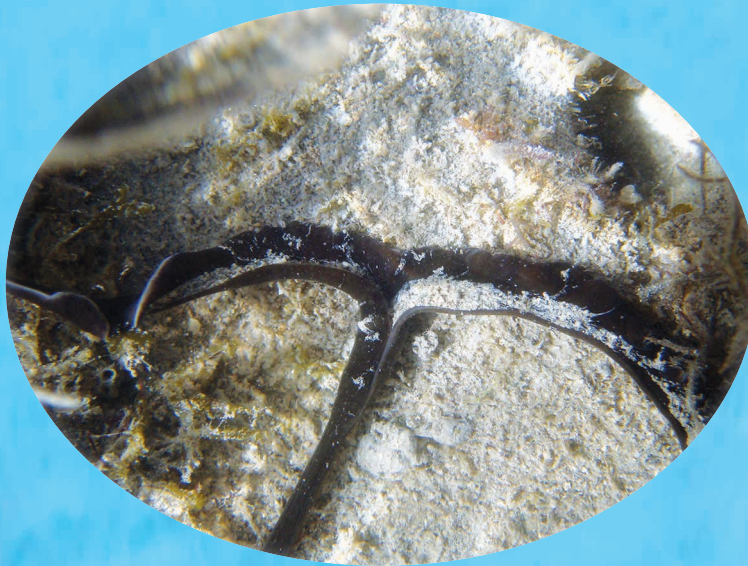




J.S. Yogesh Kumar

Zoological Survey of India, Sunderban Regional Centre, Canning-743329, West Bengal
Email: yogeshkumar.js@zsi.gov.in

ABSTRACT

A total 16 species belong to 6 genera, 2 families were reported among them 15 species from Kavaratti Island and three species from Minicoy Island. A maximum of species reported under the family Thalassematinae (15 species) and only one species reported under the family Echiurus.

Keywords : Spoon worms, Annelida, Laksadhweep, Islands

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INTRODUCTION

Echiura are the least studied group of marine animals, belong to the phylum Annelida, class polychaeta. Echiura are distributed from shallow to abyssal depths and also reported from estuaries (Walls and Jerry, 1982). So far 230 species have been described around the world (Zhang, 2011) and only 47 species were reported from Indian water. The minor group of animals commonly called spoon worms, innkeeper worms or sausage shaped worms and unsegmented bilaterally symmetrical coelomate worms, composed of two parts trunk and proboscis. Mostly, the trunk is sub-cylindrical or sac-like and muscular, bifid, sensory proboscis is attached on the anterior end of the trunk. The green colour toxic pigments present in many echiura species like *Bonellia* sp. Mostly these small animals live in soft sediments, rock crevices as well as under the boulders and deepwater ecosystem. They have a hydrostatic skeleton in large coelom and support them to move and burrow in sand, mud, soil, detritus, rubbles and rocks (Struck *et al.*, 2007; Dunn *et al.*, 2008; Struck *et al.*, 2011). The present study has supported the understanding of the Echiura composition, diversity and distribution in Lakshadweep Islands.

REVIEW OF LITERATURE

Pallas, (1766), the species first described in the name of *Echiurus (Lumbricus) echiurus* (Pallas, 1766) and placed in Phylum Annelida. After this, Stephen (1972) proposed a separate phylum Echiura. Zhang, (2011), reported 236 species belong to 5 families, 2 suborder, 2 orders. Recently valid species have been classified as per the WORMS, while the subclass Echiura were reported 195 species belong to 42 genera, 5 families and 2 suborders from elsewhere. In India, a total of 43 species were reported belonging to 13 genera from Indian water (Venkataraman *et al.*, 2015). Recently, Kumar (2020) updated the list with 47 species under 10 genera and 4 families.

A detailed literature review revealed surprisingly few studies dealing in Lakshadweep Island. Shipley's (1902) have first reported two species of Echiuran namely *Thalassema diaphanes* Sluiter, 1888 and *Anelassorhynchus moebii* (Greeff, 1879) from Minicoy Island, Lakshadweep. After these, Prasad (1935) identified *Thalassema diaphanes* Sluiter, 1888 from Indian Museum collection, Calcutta followed by Dattagupta and Menon (1965) reported *Anelassorhynchus moebii* from Minicoy Island and 7 species (*Echiurus echiurus* Pallas, 1766; *Thalassema diaphanes* Sluiter, 1888; *Anelassorhynchus inansensis* (Ikeda, 1904); *Ochetostoma palense* (Ikeda, 1924), *Ochetostoma capense* Jones and Stephen, 1955; *Ochetostoma stuhlmanni* Fischer, 1892) were reported from Kavaratti and Minicoy Island by Haldar and Dattagupta, (1991). The collections and identifications deposited in Zoological Survey of India, Calcutta.

FAUNAL DIVERSITY

In Lakshadweep Island, A total 16 species belong to 6 genera, 2 families were reported among them 15 species from Kavaratti Island and three species from Minicoy Island. A maximum of species reported under the family Thalassematinae (15 sp.) and one species reported under the family Echiurus (Table 1). The detailed systematic followed

Table 1. Status of Echiura from Lakshadweep Islands.

S. No.	Species	Kavaratti	Kiltan	Minicoy	References
	Phylum ANNELIDA				
	Class ECHIURIDA				
	Order ECHIUROIDEA				
	Family ECHIURIDAE				
	Genus Echiurus Guerin-Meneville, 1831			+	5,4
1.	<i>Echiurus echiurus</i> (Pallas, 1766)				
	Family THALASSEMATINAE				
	Genus Anelassorhynchus				
2.	<i>Anelassorhynchus inansensis</i> (Ikeda, 1904)	+			4

S. No.	Species	Kavaratti	Kiltan	Minicoy	References
3.	<i>Anelassorhynchus moebii</i> (Greeff, 1879)	+		+	2,7
	Genus Ochetostoma Leuckart & Ruppell, 1828	+			1
4.	<i>Ochetostoma caudex</i> (Lampert, 1883)				
5.	<i>Ochetostoma decameron</i> (Lanchester, 1905)	+			6
6.	<i>Ochetostoma griffini</i> (Wharton, 1913)	+			3
7.	<i>Ochetostoma hornell</i> (Prashad, 1921)	+			3,6
8.	<i>Ochetostoma kokotoniense</i> (Fischer, 1892)	+			3
9.	<i>Ochetostoma manjuyodense</i> (Ikeda, 1905)	+			3
10.	<i>Ochetostoma palense</i> (Ikeda, 1924)	+			4,5
11.	<i>Ochetostoma septemyotum</i> Datta Gupta Menon & Johnson, 1963	+			1,5
12.	<i>Ochetostoma stuhlmanni</i> (Fischer, 1892)	+			4,5
	Genus Listriolobus Fischer, 1926	+			5
13.	<i>Listriolobus brevirostris</i> Chen & Yeh, 1958				
14.	<i>Listriolobus capensis</i> (Jones & Stephen, 1955)	+			4,5
	Genus Thalassema Pallas, 1774	+		+	4,7
15.	<i>Thalassema diaphanes</i> Sluiter, 1888				
	Genus Eubonellia Fisher, 1946	+			4,5
16.	<i>Eubonellia valida</i> Fisher, 1946				
Total		15		3	

Ref. 1-Datta Gupta *et al.*, 1963; 2- Dattagupta & Menon (1965); 3-Dattagupta & Menon (1971);
4-Haldar and Dattagupta, (1991); 5-Ramlall Biseswar (2010); 6-Prasad (1935); 7- Shipley's (1902)

DISTRIBUTION

Detailed taxonomic and distribution studies of echiura are scanty in India and first study initiated from Lakshadweep Island with two species (Shipley's, 1902). Followed by very few studies reported some of species from different coast of India, East coast of India (Prasad, 1919; 1920; Dattagupta *et al.*, 1963; Halder, 1999), West Coast of India (Prasad and Awati, 1929; Dattagupta *et al.*, 1963; Dattagupta, 1986; Subba and Sastry, 2005), Andaman and Nicobar Islands (Prasad, 1935; Dattagupta and Menon, 1967). The present study confirmed echiura distribution in Lakshadweep Islands. While more detailed studies are required for exploration and assess the status from shallow to deep sea ecosystem of India.

GAP AREAS

Lakshadweep Island is a group of 36 coral islands and 10 islands are inhabited. Present study only covered two Islands during 1902 to 1991 and after no detailed studies under this group. This study partially confirmed some of Echiura species from Lakshadweep Islands. A distribution of Echiura is an indication of the need for further detailed study and exploration. The detailed study will determine the true range of distribution and understand the factors which affect the distribution.

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